Skills Summary

Adding, Subtracting, and Multiplying Polynomials

Use this reference while completing your worksheet or when studying.

Skill 1 — Adding polynomials by like terms

Rule: Only *like* terms combine — same variable, same exponent. Adding them adds the coefficients and leaves the exponent alone, because $ax^n + bx^n = (a+b)x^n$. Write both polynomials in descending order and stack the matching powers in columns.

Example: $(2x^3 + 5x - 1) + (4x^3 - 3x + 6)$
Step 1. Both are already in descending order, so I can add column by column instead of hunting for partners.
Step 2. x^3: $2 + 4 = 6$. x: $5 - 3 = 2$. constant: $-1 + 6 = 5$. $\Rightarrow$ $\boldsymbol{6x^3 + 2x + 5}$

Try it: $(x^4 - 2x^2 + 3) + (3x^4 + 2x^2 - 8)$

check: $\boldsymbol{4x^4 - 5}$

Skill 2 — Subtracting: distribute the minus first

Rule: $A - B$ means $A + (-1)B$, so the minus sign reaches **every** term of the second polynomial — including the ones already negative, which become positive. Rewrite the second polynomial with all signs flipped, then add as in Skill 1.

Example: $(5x^3 - 2x + 4) - (2x^3 + x - 9)$
Step 1. The parentheses carry a minus, so before combining anything I rewrite the second polynomial with every sign flipped.
Step 2. It becomes $-2x^3 - x + 9$; now add: $5 - 2 = 3$ on x^3, $-2 - 1 = -3$ on x, $4 + 9 = 13$ on the constant. $\Rightarrow$ $\boldsymbol{3x^3 - 3x + 13}$

Try it: $(4x^4 + x^2 - 6) - (x^4 - 3x^2 + 2)$

check: $\boldsymbol{3x^4 + 4x^2 - 8}$

Skill 3 — Multiplying with an area box

Rule: Give the first factor's terms the rows and the second factor's terms the columns. Each cell is one product (multiply coefficients, add exponents); the answer is the sum of all the cells. An m-term factor times an n-term factor always fills $m \times n$ cells — if you have fewer, you skipped one.

Example: $(x + 4)(x^2 - 3x + 2)$
Step 1. Two terms times three terms means six products, so the box keeps me from losing any of them.
Step 2. Row x: x^3, $-3x^2$, $2x$. Row 4: $4x^2$, $-12x$, 8. Collect: $-3x^2 + 4x^2 = x^2$ and $2x - 12x = -10x$. $\Rightarrow$ $\boldsymbol{x^3 + x^2 - 10x + 8}$

Try it: $(x-2)(x^2+5x-1)$

check: $\boldsymbol{x^3 + 3x^2 - 11x + 2}$

Skill 4 — Checking a product two ways

Rule: Two independent checks:

- **Substitute** a value into the factored form and the expanded form. Equal polynomials agree at every input, so any disagreement proves an error.
- **Factor back**. If the expansion factors to what you started with, the arithmetic held.

Example: Does $(x+4)(x^2-3x+2) = x^3 + x^2 - 10x + 8$? Test $x = 3$.
Step 1. One substitution cannot prove equality, but it can disprove it cheaply, so I try a value that is easy to compute and not a root.
Step 2. Factored: $(7)(9-9+2) = 14$. Expanded: $27 + 9 - 30 + 8$. $\Rightarrow$ both give **14**, so the expansion survives the test

Try it: Factor $x^3 - x^2 - 6x$ back to a product of three factors.

check: $\boldsymbol{x(x-3)(x+2)}$